

Surface and Interfacial Tension Effects on the Wrinkling of Multilayer Dielectric Elastomer Films

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Abstract

Multilayer dielectric-elastomer films—a stiff dielectric skin bonded to a softer compliant substrate, or laminates of several dissimilar layers—are increasingly used in soft actuators and sensors, and when operated in fluidic environments they experience surface tension not only on their outer surface but also, through the mismatch of interfacial energies, on the buried interfaces between layers. Here we extend the elastocapillary wrinkling analysis of compressed dielectric elastomers to multilayer laminates. Building on the single-layer Young–Laplace framework, we formulate the bifurcation problem for a bonded bilayer carrying a free-surface tension and an independent interfacial tension, and generalize it to an arbitrary N -layer stack. Interfacial surface energy enters as a Laplace-pressure jump in the normal traction across the interface, raising the critical strain and coarsening the selected wavelength much as free-surface tension does. For laminates of three or more layers a qualitatively new effect emerges: a sufficiently large free-surface tension flattens the outer surface while a buried stiff layer continues to buckle, so that surface tension selects which interface fails first. The predictions are verified with nonlinear finite-element simulations in which the same edge-based Young–Laplace surface stress is applied on internal interfaces as on free surfaces.

Keywords: Wrinkling Instability, Elastocapillary, Interfacial Energy, Multilayer Films, Perturbation Analysis

1. Introduction

Because soft materials like gels and elastomers often operate under states of compression, there have been significant efforts to examine their mechanical stability under such loading (Biot, 1963; Gent and Cho, 1999; Biot and Romain, 1965; Tanaka et al., 1987; Gent, 2005; Hong et al., 2009). Recently, efforts have focused on examining the stability of surfaces due to the formation of instabilities such as wrinkles and creases under compression (Cao and Hutchinson, 2012; Li et al., 2012; Jin et al., 2014; Cai et al., 2012; Hong and Gao, 2013; Weiss et al., 2013). The studies of wrinkling instabilities have typically been analytical in nature, based on linear perturbation analysis, which enables predictions of critical strains and instability wavelengths in these soft structures. These studies have also been insightfully combined with experiments (Cai et al., 2010, 2012; Jin et al., 2015), and also numerical (finite element) analyses (Cao and Hutchinson, 2012; Jin and Suo, 2015; Jin et al., 2015; Wang and Zhao, 2014, 2016).

More recently, linear perturbation analyses have been applied to an interesting class of soft, active materials - dielectric elastomers (DEs). These materials have drawn significant interest due to the large deformations they can undergo resulting from applied electrostatic loading (Carpi et al., 2010; Brochu and Pei, 2010; Biddiss and Chau, 2008; Mirfakhrai et al., 2007). Furthermore, DEs often form surface instabilities like creasing, wrinkling and cratering, which have been studied via experiment, theory and computation (Suo, 2010; Plante and Dubowsky, 2006; Zhou et al., 2008; Wang et al., 2012; Shivapooja et al., 2013; Wang et al.,

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2011; Park et al., 2012, 2013), and also using analytic techniques based on stability analyses (Zhao and Suo, 2007, 2009) and the nonlinear field theory of Suo et al. (2008).

While DEs have drawn significant interest in recent years, many promising technological innovations, such as generating electromechanical motion during magnetic resonance imaging (Carpi et al., 2008), operating soft, underwater robots (Rus and Tolley, 2015; Kim et al., 2013), manipulating microfluidic flow (Holmes et al., 2013), focusing a tunable lens (Carpi et al., 2011), the preparation of bioinspired surfaces (Shivapooja et al., 2013), underwater grippers (Laschi et al., 2012) and soft body locomotion (Marchese et al., 2014), involve operation of the DE in fluidic environments, where elastocapillary effects due to surface tension can become prominent (Roman and Bico, 2010; Liu and Feng, 2012). While prior works have examined either the effects of surface tension alone on surface instabilities in elastomers (Chen et al., 2012) and hydrogels (Kang and Huang, 2010), or the coupled effects of surface tension and electric fields on surface instabilities in constrained DE films (Seifi and Park, 2016; Wang et al., 2016), a general stability analysis of compressed DE films subject to both surface tension and electric fields has not been performed.

The objective of this work is to extend the elastocapillary analysis of surface instabilities in DEs from a single layer to *multilayer* laminates, in which surface energy acts both on the free outer surface and on the buried interfaces between dissimilar layers. Building on the single-layer Young–Laplace framework, we first summarize the perturbation analysis of a compressed single film, then formulate the bifurcation problem for a bonded bilayer with an independent interfacial tension, generalize it to an N -layer stack, and verify the predictions with nonlinear finite-element simulations in which the same Young–Laplace surface stress is applied on internal interfaces.

2. Governing Equations and Incremental Forms

We first present, for a DE film subject to surface tension and under compressive loading, the governing equations and their incremental forms. We assume that the surface of the film is subject to Young–Laplace boundary conditions in order to account for elastocapillary effects on the surface (Saksono and Peric, 2006). We restrict attention to mechanical (compressive) loading throughout; the electrostatic actuation of these films is treated in the companion single-layer study. The perturbation analysis we utilize is based on those performed by Cai et al. (2010); Kang and Huang (2010); Jin et al. (2014, 2015)

2.1. Dielectric Elastomer Subject to Surface Tension

We first consider a DE film under compression, that is also subject to surface tension effects on its top surface, as illustrated in Figure (1). The deformation of the body is described by the field $\mathbf{x}(\mathbf{X})$. The deformation gradient tensor $\mathbf{F}$ is defined as:

$$\mathbf{F} = \nabla \mathbf{x} \quad (1)$$

The DE is modeled as an incompressible neo-Hookean material with the free energy density function:

$$W(\mathbf{F}) = \frac{\mu}{2} \mathbf{F} : \mathbf{F} - \pi(\det \mathbf{F} - 1) \quad (2)$$

where $\mathbf{A} : \mathbf{B} = \text{tr}(\mathbf{A}^T \mathbf{B})$ denotes the tensor inner product. The first Piola–Kirchhoff (nominal) stress tensor $\mathbf{S}$ can be derived as:

$$\mathbf{S} = \frac{\partial W(\mathbf{F})}{\partial \mathbf{F}} = \mu \mathbf{F} - \pi \mathbf{F}^{-T} \quad (3)$$

where μ is the shear modulus, and $\pi = \pi(\mathbf{X})$ is the Lagrange multiplier to enforce the constraint of incompressibility: $J = \det \mathbf{F} = 1$. The nominal stress satisfies the reference equilibrium equation:

$$\nabla \cdot \mathbf{S} = \mathbf{0} \quad (4)$$

On the surface of the body the film is subjected to the Young–Laplace boundary condition (Saksono and Peric, 2006) with surface tension γ :

$$\mathbf{S} \mathbf{n} dA = 2\gamma \kappa \mathbf{n} da \quad (5)$$

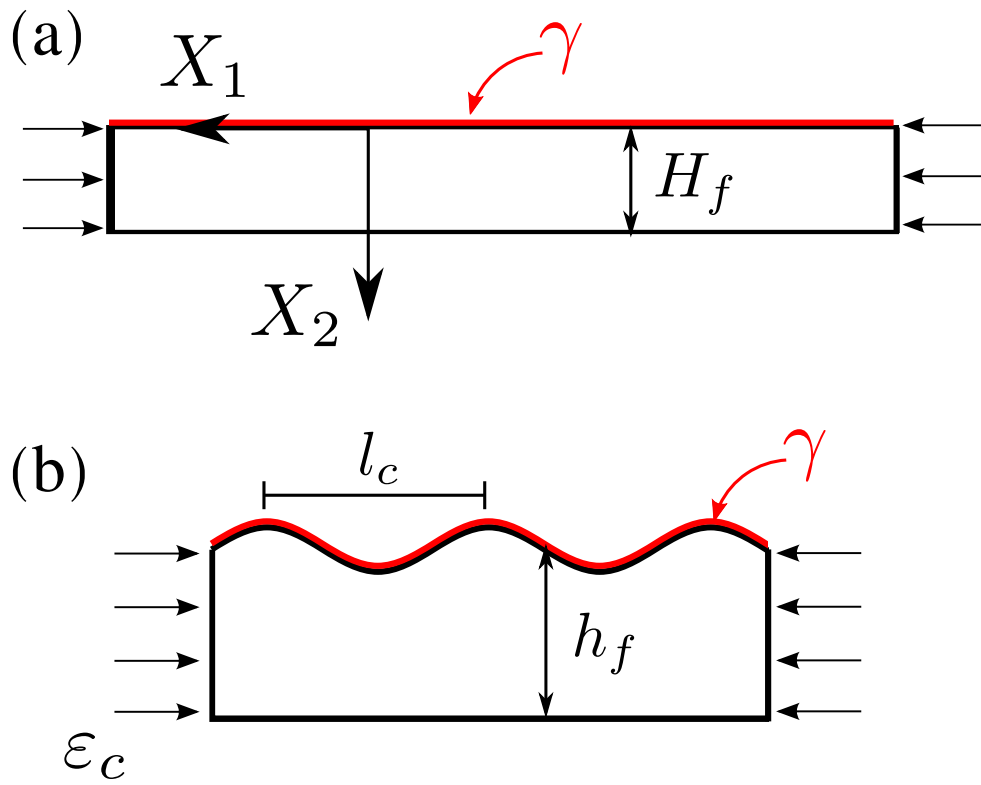


Figure 1: A perfectly flat film subject to compression and surface tension γ . (a) Undeformed configuration; (b) the formation of wrinkles with wavelength l_c .

where $\mathbf{N}$ is the unit normal vector to the surface of the body in the undeformed state, $\mathbf{n}$ is the unit normal vector in the current state, and κ is the mean curvature of the surface.

We now add a perturbation into the state of finite deformation with position $\mathbf{x}_0(\mathbf{X})$, deformation gradient $\mathbf{F}_0(\mathbf{X})$ and nominal stress $\mathbf{S}_0(\mathbf{X})$. If the deformation in the z -direction is constrained, only 2D plane strain deformation is considered. The background deformation gradient of this state is

$$\mathbf{F}_0 = \begin{bmatrix} \lambda & 0 \\ 0 & 1/\lambda \end{bmatrix} \quad (6)$$

Therefore the corresponding base position is $\mathbf{x}_0 = \mathbf{F}_0 \mathbf{X}$. Now by adding a small displacement perturbation $\delta \mathbf{x}$ and pressure perturbation $\delta \pi$, we write the perturbed state as $\mathbf{x} = \mathbf{x}_0(\mathbf{X}) + \delta \mathbf{x}(\mathbf{X})$ and $\pi = \pi_0 + \delta \pi$. Both the base state and the perturbed state satisfy the same governing equations in (1) and (4). The incremental form of the deformation gradient reduces to

$$\delta \mathbf{F} = \nabla(\delta \mathbf{x}) \quad (7)$$

and the equilibrium equation to

$$\nabla \cdot (\delta \mathbf{S}) = \mathbf{0} \quad (8)$$

where the incremental nominal stress tensor can be obtained using directional differentiation at $\mathbf{F}_0$ of the stress in (3):

$$\delta \mathbf{S} = \mu \delta \mathbf{F} - \delta \pi \mathbf{F}_0^{-T} + \pi_0 \mathbf{F}_0^{-T} (\delta \mathbf{F})^T \mathbf{F}_0^{-T} \quad (9)$$

Inserting (9) into (8) we can obtain the following equation:

$$\mu \nabla^2(\delta \mathbf{x}) + \pi_0 \mathbf{F}_0^{-T} \nabla \cdot [(\delta \mathbf{F})^T \mathbf{F}_0^{-T}] - \mathbf{F}_0^{-T} \nabla(\delta \pi) = \mathbf{0} \quad (10)$$

Taking the directional derivative of the incompressibility condition yields:

$$\mathbf{F}_0^{-T} : \delta \mathbf{F} = 0 \implies \frac{1}{\lambda} \frac{\partial(\delta x_1)}{\partial X_1} + \lambda \frac{\partial(\delta x_2)}{\partial X_2} = 0 \quad (11)$$

where the perturbation of the boundary condition reads:

$$\delta \mathbf{S} \mathbf{N} = 2\gamma \delta \kappa \mathbf{F}_0^{-T} \mathbf{N} \quad (12)$$

where the baseline flat state satisfies $\kappa = 0$, so the surface tension contributes only through the curvature increment $\delta \kappa$. The perturbed surface lives in the *current* configuration, where the in-plane coordinate is $x_1 = \lambda X_1$; its curvature increment is therefore

$$\delta \kappa = -\frac{\partial^2(\delta x_2)}{\partial x_1^2} = -\frac{1}{\lambda^2} \frac{\partial^2(\delta x_2)}{\partial X_1^2} = \frac{K^2}{\lambda^2} f_2(X_2) \cos(KX_1), \quad (13)$$

the factor λ^{-2} arising because the wavelength is stretched by λ . The Young–Laplace force $\gamma \delta \kappa$ acts per unit *current* surface area; mapping it back to a nominal traction over the reference surface introduces the surface stretch (Jacobian) $da/dA = \lambda$, so the surface tension enters the normal traction balance as the restoring, stabilizing term $(\gamma K^2/\lambda) \delta x_2$. (For a strain-dependent, nominal surface stress the factor would instead be $1/\lambda$ alone; the two coincide here because γ is the physical residual tension.) At the bottom of the film ($X_2 = H_f$) we impose a constrained-sliding (frictionless roller) base, $t_1 = 0$ and $f_2 = 0$, matching the substrate condition of the finite-element model; a bonded base ($f_1 = f_2 = 0$) shifts the onset strain by less than 0.01 and is not pursued here:

$$\delta S_{12} = 0 \quad \text{and} \quad \delta x_2 = 0 \quad (14)$$

3. Incremental perturbation framework

To determine the onset of wrinkling in a compressed film, we look for non-trivial solutions to the eigenvalue problem via Fourier representation:

$$\begin{aligned}\delta x_1(X_1, X_2) &= f_1(X_2) \sin(KX_1) \\ \delta x_2(X_1, X_2) &= f_2(X_2) \cos(KX_1) \\ \delta \pi(X_1, X_2) &= f_3(X_2) \cos(KX_1)\end{aligned}\tag{15}$$

Substituting these fields into the mechanical equilibrium lines and eliminating the pressure envelope $f_3(X_2)$ yields the classical fourth-order ordinary differential equation for the vertical envelope function f_2 :

$$\frac{d^4 f_2}{dX_2^4} - K^2 (\lambda^{-4} + 1) \frac{d^2 f_2}{dX_2^2} + K^4 \lambda^{-4} f_2 = 0\tag{16}$$

where $K = 2\pi/L$ is the reference wavenumber. Within each layer f_2 is therefore a combination of the four exponential modes $\exp(\pm KX_2)$ and $\exp(\pm KX_2/\lambda^2)$, and the incompressibility constraint gives $f_1 = -(\lambda^2/K) f_2'$. The boundary and interface conditions that close the eigenvalue problem are assembled directly for the multilayer laminate in the next section.

4. Multilayer films with interfacial surface energy

Many soft active structures are not monolithic but layered—a stiffer dielectric skin bonded to a softer compliant substrate, or two dielectric layers of differing stiffness joined at a common electrode. When such a structure is fabricated or operated in a fluidic environment, surface energy is present not only on the exposed free surface but also on the buried *interface* between the two materials. We now extend the perturbation analysis of Section 3 to a compressed two-layer film, retaining a distinct surface tension on the top surface, γ_t , and on the internal interface, γ_i .

4.1. Bilayer perturbation problem

Consider two bonded incompressible neo-Hookean layers resting on a rigid base at $X_2 = 0$. Layer 1 (substrate, shear modulus μ_1) occupies $0 \leq X_2 \leq H_1$ and layer 2 (film, shear modulus μ_2) occupies $H_1 \leq X_2 \leq H_1 + H_2$, with the interface at $X_2 = H_1$. Because the layers are bonded and the base imposes a uniform in-plane compression, both share the common background stretch $\mathbf{F}_0 = \text{diag}(\lambda, 1/\lambda)$; the traction-free top and flat interface fix the per-layer background pressure $\pi_0^{(k)} = \mu_k/\lambda^2$.

Each layer admits the same incremental field structure as the single-layer problem (15), so within layer k the vertical envelope $f_2^{(k)}$ satisfies the fourth-order equation (16) and is a combination of four exponential modes $\exp(\pm KX_2)$ and $\exp(\pm KX_2/\lambda^2)$ —eight unknown amplitudes in total. The incompressibility constraint gives $f_1 = -(\lambda^2/K) f_2'$, and eliminating the pressure through the horizontal equilibrium equation yields $f_3 = \mu\lambda^3(f_2'''/K^2 - f_2')$. The incremental nominal tractions on a horizontal face then reduce to

$$t_1 = -\mu \left[\frac{\lambda^2}{K} f_2'' + \frac{K}{\lambda^2} f_2 \right], \quad t_2 = \mu \left[(2 + \lambda^4) f_2' - \frac{\lambda^4}{K^2} f_2''' \right],\tag{17}$$

where t_1 and t_2 are the incremental shear and normal tractions.

4.2. Boundary and interface conditions

The eight amplitudes are fixed by eight conditions. At the rigid base the displacement is clamped,

$$f_1^{(1)}(0) = 0, \quad f_2^{(1)}(0) = 0.\tag{18}$$

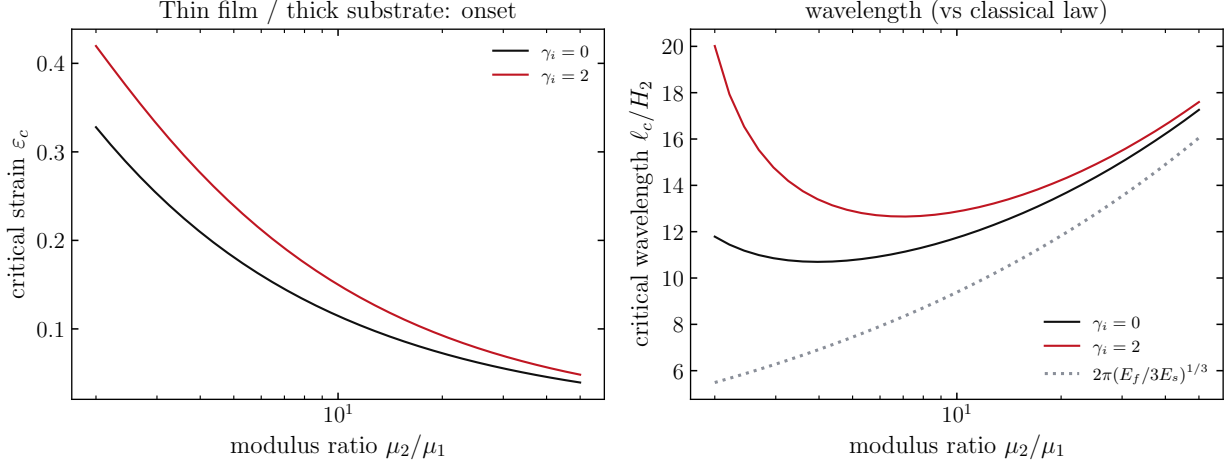


Figure 2: Bilayer instability with interfacial surface energy. (a) Critical strain ε_c versus modulus ratio μ_2/μ_1 for interfacial tension $\gamma_i = 0$ (solid) and $\gamma_i > 0$ (dashed); (b) the corresponding selected wavelength ℓ_c/H_2 . Curves are the bilayer perturbation prediction (21) for a thin film on a thick substrate ($H_1 = 20$, $H_2 = 1$); the dotted line in (b) is the classical film/substrate law $2\pi(E_f/3E_s)^{1/3}$.

At the bonded interface $X_2 = H_1$ the displacement and the tangential traction are continuous, while the *normal* traction jumps by the Laplace pressure of the interfacial tension,

$$f_1^{(1)} = f_1^{(2)}, \quad f_2^{(1)} = f_2^{(2)}, \quad t_1^{(1)} = t_1^{(2)}, \quad t_2^{(2)} - t_2^{(1)} = \frac{\gamma_i K^2}{\lambda} f_2. \quad (19)$$

The jump condition (19) is the new ingredient: a tensioned interface, like a stretched membrane, resists curvature and therefore couples the two layers through a restoring normal traction proportional to γ_i . Finally, at the free top surface $X_2 = H_1 + H_2$,

$$t_1^{(2)} = 0, \quad t_2^{(2)} + \frac{\gamma_t K^2}{\lambda} f_2^{(2)} = 0, \quad (20)$$

which is the bilayer counterpart of the single-layer Young–Laplace condition (12). Collecting the eight conditions into the homogeneous system $\mathbf{M}\mathbf{c} = \mathbf{0}$, non-trivial solutions require

$$\det \mathbf{M}(\lambda, K; \mu_2/\mu_1, H_1, H_2, \gamma_t, \gamma_i) = 0, \quad (21)$$

and the critical compressive strain $\varepsilon_c = 1 - \lambda_c$ follows from minimizing over the wavenumber K .

Two limits verify (21). For a homogeneous film ($\mu_2 = \mu_1$) with $\gamma_t = \gamma_i = 0$ the determinant reproduces Biot’s surface threshold $\varepsilon_c \approx 0.46$. With $\gamma_i = 0$ and a stiffness contrast, it recovers the classical stiff-film/compliant-substrate wrinkling wavelength $\ell_c \approx 2\pi H_2 (E_f/3E_s)^{1/3}$ to within the finite-substrate correction. Switching on the interfacial tension γ_i raises the critical strain and lengthens the selected wavelength (Figure 2): the buried interface, like the free surface, penalizes short-wavelength undulation, so the two surface energies act in concert to delay and coarsen the instability.

We verify (21) against nonlinear finite-element calculations in which the two layers are meshed as bonded element blocks sharing a conforming row of interface nodes, and the surface traction is applied on both the top side set and the internal interface side set. The laminate ($L_f = 40$, substrate $H_1 = 8$, film $H_2 = 1$) is compressed laterally under displacement control until the top-surface undulation reaches the bifurcation, and the onset strain and wavelength are extracted at each modulus ratio. As shown in Figure 3, the FE onset strains track the perturbation prediction across $5 \leq \mu_2/\mu_1 \leq 20$, decreasing monotonically from $\varepsilon_c \approx 0.18$ to 0.08, with the FE values lying a uniform $\sim 10\%$ above the linear theory—the expected offset between an infinitesimal-amplitude bifurcation and a finite-amplitude onset measured at a small but nonzero amplitude.

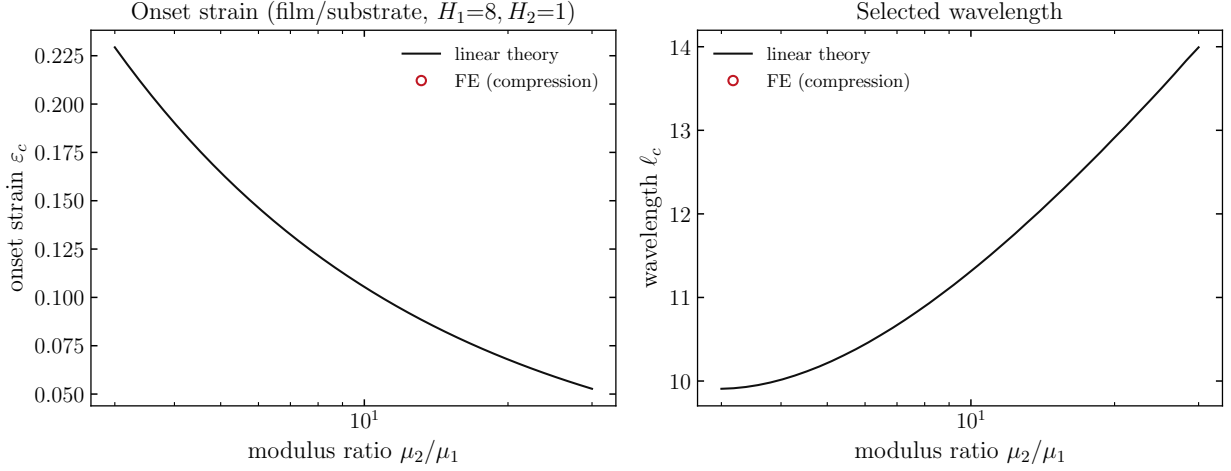


Figure 3: Finite-element verification of the bilayer onset criterion (21) for the film/substrate geometry ($H_1 = 8$, $H_2 = 1$, $\gamma_i = 0$). (a) Onset compressive strain ε_c and (b) selected wavelength ℓ_c versus modulus ratio: solid line, linear stability theory; markers, nonlinear FE under lateral compression. The FE onset is measured where the top-surface undulation first reaches 10% of the film thickness, which accounts for the small uniform upward offset relative to the infinitesimal-amplitude theory.

The selected wavelengths likewise agree to within the discrete set permitted by the finite specimen length. Most importantly, switching on the interfacial tension at fixed modulus ratio ($\mu_2/\mu_1 = 10$) raises the onset strain from $\varepsilon_c = 0.118$ to 0.213 in the FE calculation, in close agreement with the corresponding shift $0.105 \rightarrow 0.189$ predicted by (21). This confirms that interfacial surface energy is a distinct and quantitatively accessible control parameter for the surface morphology of layered dielectric elastomers.

4.3. Generalization to an N -layer laminate

The bilayer construction extends directly to an arbitrary stack of N bonded incompressible neo-Hookean layers, with shear moduli μ_k and thicknesses H_k ($k = 1, \dots, N$, ordered from the rigid base upward) and an independent surface tension γ_i on each internal interface ($i = 1, \dots, N - 1$) in addition to the free-surface tension γ_t . Since the bonded stack shares a single in-plane stretch λ , every layer carries the four-mode envelope of Section 4 in its own local coordinate, giving $4N$ unknown amplitudes. These are fixed by $4N$ conditions—two at the clamped base, the four continuity/Laplace conditions (19) at each of the $N - 1$ interfaces, and two at the free top—assembled into a banded $4N \times 4N$ matrix $\mathbf{M}(\lambda, K)$ whose vanishing determinant is the bifurcation condition. The assembly is the standard transfer-matrix (propagator) form and is linear in N , so laminates with many layers are solved at negligible cost; the null vector of $\mathbf{M}$ at threshold yields the mode shape, which localizes the incipient instability to a particular surface or interface.

This generalization exposes a qualitatively new effect that the single free surface cannot display. Consider a thin stiff sheet buried between two soft layers. When the free-surface tension is small, compression triggers a buckling mode of the stiff sheet that also deflects the top surface. As γ_t is increased, the free-surface undulation is progressively penalized and flattened, yet the buried sheet continues to buckle at essentially the same critical strain, because its mode is localized away from the free surface and is therefore insensitive to γ_t . Beyond a threshold surface tension the buried interface buckles *first*, with the top surface remaining flat (Figure 4): surface tension does not merely shift the onset of a single instability but can *select which interface fails first*, decoupling internal pattern formation from the external surface. The same $4N \times 4N$ framework provides the forward map for the inverse design of layered actuators with a prescribed buried or surface morphology; because each evaluation is inexpensive, it is also a natural basis for a reduced-order surrogate over the layer parameters (μ_k, H_k, γ_i) when embedded in a design-optimization loop.

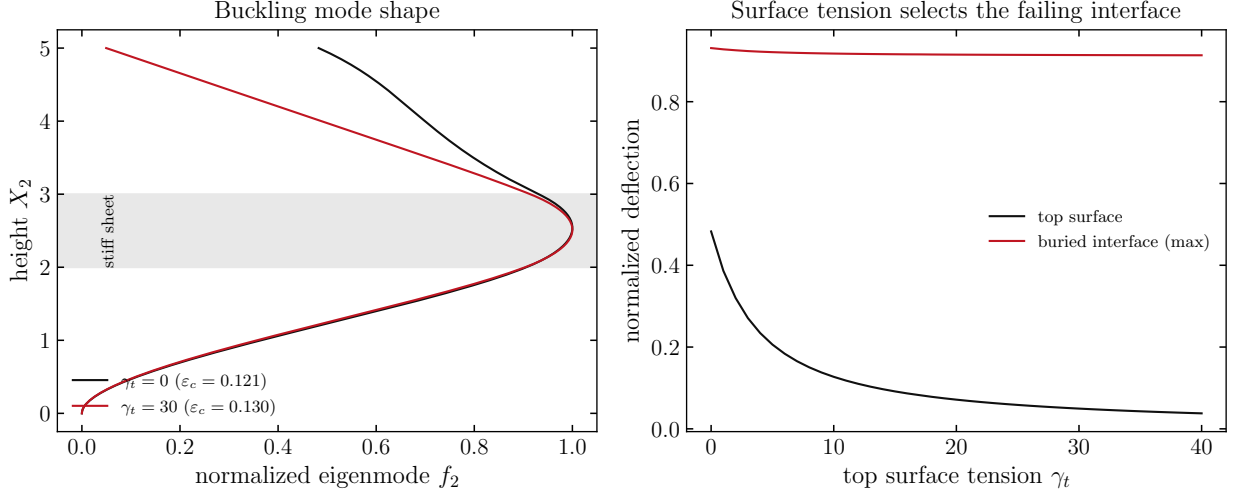


Figure 4: Surface tension selects the failing interface in a three-layer soft/stiff/soft laminate. (a) At low free-surface tension the buried stiff sheet buckles and the top surface follows; (b) at high free-surface tension the top surface is flattened while the buried sheet still buckles at nearly the same critical strain, so the interface instability preempts the surface one. Curves are the normalized vertical-displacement eigenmode $f_2(X_2)$ from the N -layer determinant.

5. Conclusions

We have extended the elastocapillary wrinkling analysis of compressed dielectric elastomers from a single film to multilayer laminates, and verified it with nonlinear finite-element calculations. Surface energy on a buried interface between dissimilar dielectric layers enters the bifurcation problem as a Laplace-pressure jump in the normal traction (19), raising the critical strain and coarsening the selected wavelength much as free-surface tension does; the bilayer determinant reduces correctly to the homogeneous Biot limit and to the classical stiff-film/soft-substrate wrinkling law. For laminates of three or more layers a qualitatively new effect emerges: a sufficiently large free-surface tension flattens the outer surface while a buried stiff layer continues to buckle, so that surface tension selects which interface fails first. On the computational side, the same edge-based Young–Laplace surface stress is applied on internal interfaces exactly as on free surfaces, by declaring the interface as an element side set. Interfacial energy thus emerges as an additional, independently tunable control over the surface morphology of layered soft actuators, and the N -layer determinant provides a compact forward model for their design.

Appendix A. Finite-element application of surface tension on an internal interface

The free-surface Young–Laplace traction used here was introduced and validated for the *exterior* surface of a single body in Seifi and Park (2016). The multilayer results of Section 4 require the same surface energy on a buried *interface* between two bonded layers. We summarize how this is done without altering the surface constitutive model; only the set of faces on which it acts is extended.

Discrete surface stress. The surface energy is carried on the element edges that lie on the surface. For a deformed surface edge with end nodes at current positions $\mathbf{x}_a, \mathbf{x}_b$, let $\Delta\mathbf{x} = \mathbf{x}_a - \mathbf{x}_b$ with current length $\ell = |\Delta\mathbf{x}|$. The edge carries the constant Young–Laplace surface stress $\sigma_s = \gamma$, with γ the surface tension. The corresponding nodal forces are

$$\mathbf{f}_a = -\frac{\sigma_s}{\ell} \Delta\mathbf{x}, \quad \mathbf{f}_b = +\frac{\sigma_s}{\ell} \Delta\mathbf{x}, \quad (\text{A.1})$$

so each edge pulls its end nodes together along the deformed tangent with magnitude σ_s . At a node shared by two surface edges the net force is $\sigma_s(\hat{\mathbf{t}}_2 - \hat{\mathbf{t}}_1)$, proportional to the discrete curvature times the tension—the

discrete Laplace pressure—and the consistent geometric and material tangent contributions are assembled likewise.

Extension to an interface. On a free surface the active faces are the exterior element faces, identified automatically as those without a neighboring element. An internal interface between two element blocks is fully surrounded by material and is therefore *not* detected as a free surface. To apply surface energy there, the interface is declared as an element side set and supplied to the surface-tension input exactly as a free surface would be. The element then (i) collects the side-set faces, (ii) merges their parent elements into the surface-element list—these are otherwise absent, having no exterior face—and (iii) marks the designated faces as active. The identical edge-based stress σ_s and nodal forces are then evaluated on the interface edges. Because the two layers are meshed conformally and share the interface nodes (a bonded, displacement-continuous interface), the curvature-driven force acts on the shared nodes and is transmitted to both layers, realizing the interfacial Laplace-pressure jump of Eq. (19).

Two consequences follow. First, because the surface constitutive model is unchanged and only the face set is extended, the prior validation of the free-surface model carries over to the interface. Second, surface tensions are specified per side set, so distinct values may be assigned to the top surface and to each internal interface independently, as exploited in Section 4.3.

Appendix B. Acknowledgements

HSP and SS acknowledge funding from the ARO, grant W911NF-14-1-0022.

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